The Impacts of Salinity on Different Crab Species at Elevated Temperatures

https://www.inaturalist.org/taxa/69170-Hemigrapsus-oregonensis

https://www.exoticsguide.org/carcinus_maenas

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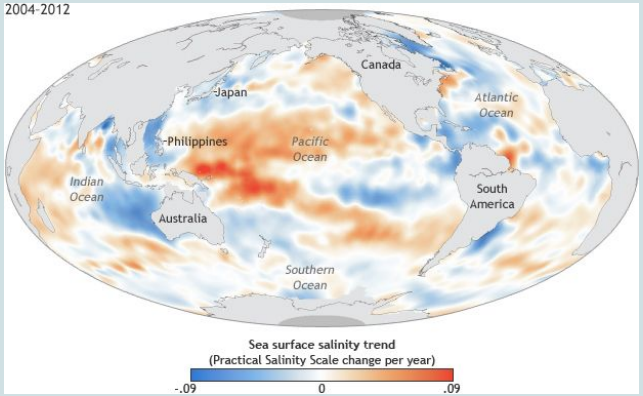
Background

- Climate change is altering temperatures and salinities (Pörtner 2012)
- Intertidal species experience rapid changes to their environment
- Increase in temperature and salinity is raising metabolic demands in ectotherms (Schiffer et al. 2014).
- Physiological tolerance is crucial for survival (Somero 2010).



Species: Hairy Shore Crab

Hemigrapsus oregonensis

https://www.blog.illustraciencia.info/2018/06/hemigrapsus-oregonensis-michelle-wu.html

https://en.wikipedia.org/wiki/History_of_the_west_coast_of_North_America

Native Range: (Cowles, 2005)

-Resurrection Bay, Alaska
-Bahia de Todos Santos, Baja

Ecological Role:

-Omnivorous Scavenger
-Major Food Source
-Green Crab Competitor

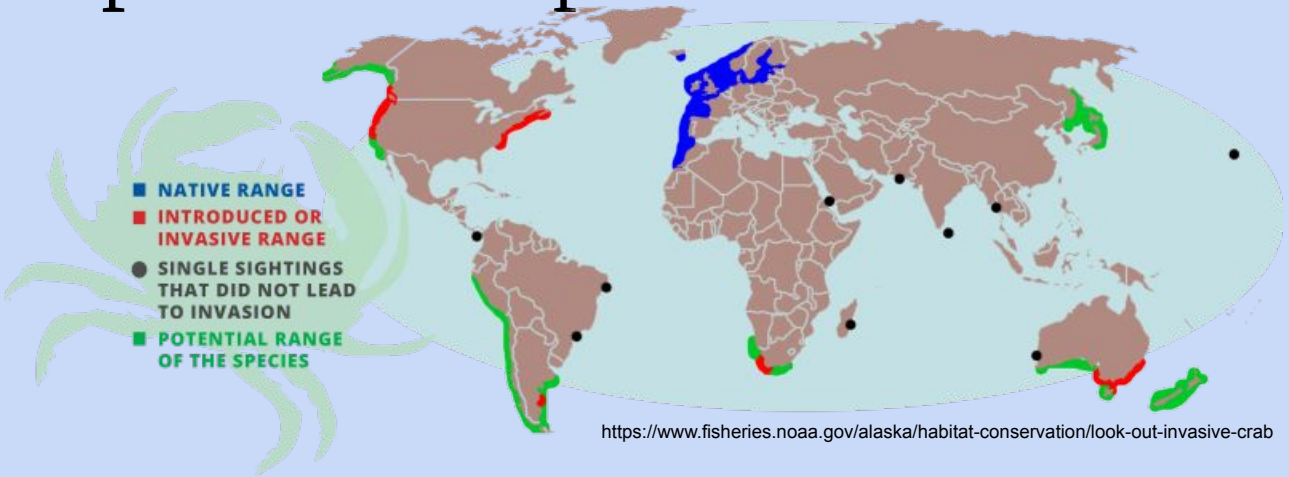
Acclimation:
(Gross, 1963)

-Osmoregulatory Acclimation
-Sudden or Gradual Acclimation

Adaptation:
(Felder and Windsor, 2020)

-Crabs found in Texas Salt Spring
-18–22 °C, 43 ppt
-Genetic match to *H. oregonensis*

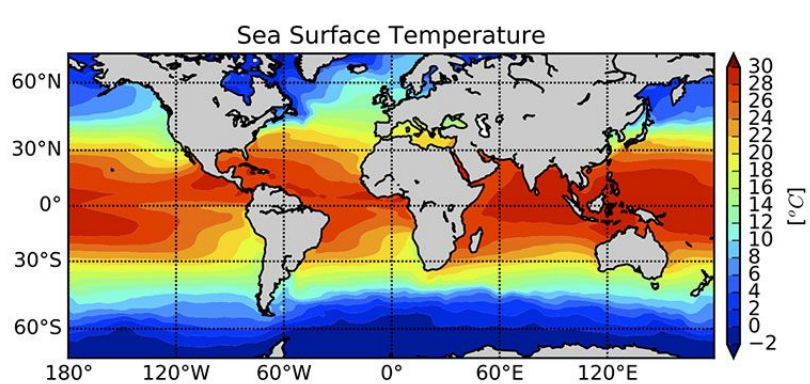
Species: European Green Crab



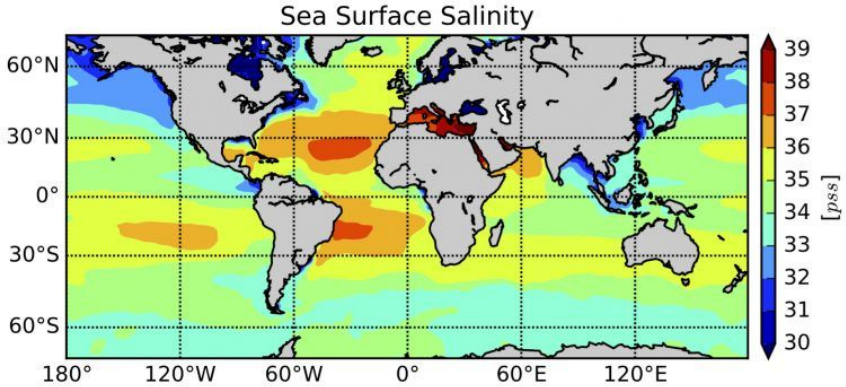
https://www.fisheries.noaa.gov/alaska/habitat-conservation/look-out-invasive-crab

Carcinus maenas

https://www.bates.edu/biology/2015/02/04/biology-seminar-series-2/



https://oceansconnectes.org/en/average-ocean-temperature-2/



https://oceansconnectes.org/en/average-ocean-salinity/

Research Question

At increased temperatures, how do the physiological and behavioral responses change in crabs (*C. maenas*, *H. oregonensis*) at different salinities?

- How does energy use change in terms of osmoregulation?
- How does behavior change in different salinities when crabs are already under abiotic stress factors?

Hypothesis

Ho: There is no change in the physiology or behavior of crabs when salinity changes at an elevated temperature

Ha: There is a change in the physiology and behavior of crabs when salinity changes at an elevated temperature

- The metabolic rate, feeding time, and O2 levels will decreases as salinity increases

Experimental Design

High Temperature

Ambient Salinity

H. Oregonensis

High Temperature

High Salinity

H. Oregonensis

High Temperature

Ambient Salinity

C. maenas

High Temperature

High Salinity

C. maenas

Data collection:

- **Total protein assays: energy use, metabolism proxy**
- **O2 respirometry: metabolic changes over time**
- **Feeding mass: foraging efficiency, proxy for growth**
- **Feeding time: behavioral stress response**

Replicates:

- **22 H. oregonensis**
- **2 C. maenas**

Repeated measurements:

- **TPA: 1x per crab at the end of study**
- **O2: 3x per crab (1 each week)**
- **Feeding time: 3x per crab (1 each week)**

Citations:

Pörtner, H. O. (2012). *Integrating climate-related stressor effects on marine organisms*. Journal of Experimental Biology.https://www.int-res.com/articles/theme/m470p273.pdf

Schiffer, M et al. (2014). *Tolerance of crustaceans to environmental stress*. Frontiers in Zoology.

Somero, G. N. (2010). *The physiology of climate change: how potentials for acclimatization and genetic adaptation will determine winners and losers*. Journal of Experimental Biology.

Cowels, D. (2005). Hemigrapsus oregonensis (Oregon shore crab).https://inverts.wallawalla.edu/Arthropoda/Crustacea/Malacostraca/Eumalacostraca/Eucarida/Decapoda/Brachyura/Family_Grapsidae/Hemigrapsus_oregonensis.html

Gross, W. (1963). Comparative Biochemistry and Physiology, Acclimation to hypersaline water in a crab.https://doi.org/10.1016/0010-406X(63)90042-2.

Felder, D. L., & Windsor, A. M. (2020). Introduced Hemigrapsus oregonensis (Dana, 1851) formerly colonized an inland Texas salt spring, as now underpinned by COI barcode sequence analysis. BioInvasions Records. https://doi.org/10.3391/bir.2020.9.2.12